

LAB 6: TEST-DRIVEN-PROMPTING

Agile Logbook: PII Redactor Middleware

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Phase 1: Failure Verification (Red State)

- **Action:** Created the `test_redactor.py` suite containing tests for standard redaction, missing data, malformed types, and immutability (Information Hiding). Executed `pytest test_redactor.py -v` before any functional logic was implemented in `redactor.py`.
- **Result:** Tests failed as expected (Red State verified). This confirmed the test suite correctly enforced the strict Interface Contract.

Phase 2: Prompt Execution (AI Pilot)

Prompt Used

I am practicing Test-Driven Development. I have written a `pytest` suite for a middleware function called `redact_pii(data)`.

The tests expect the function to:

- Redact 'email' and 'phone' keys with '[REDACTED]'.
- Handle empty dicts and missing keys.
- Handle malformed data types without throwing errors.
- Not mutate the original input dictionary.

Here is my test suite: [Pasted contents of `test_redactor.py`].

Generate the Python code for `redactor.py` that will make all of these tests pass without violating the immutability constraint.

Phase 3: Pass Verification (Green State)

- **Action:** Saved the AI-generated logic (which correctly utilized `deepcopy` and safe type-checking) into `redactor.py` and executed the test suite again.
- **Result:** All tests passed successfully, confirming the AI adhered to the Interface Contract defined by the edge case tests.

Terminal Output:

```
===== test session starts
=====
platform win32 -- Python 3.13.5, pytest-9.0.3, pluggy-1.5.0 -- C:\Users\
youse\Miniconda3\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\youse\Desktop\labsoft
plugins: anyio-4.13.0
collected 4 items

test_redactor.py::test_standard_redaction PASSED [
25%]
test_redactor.py::test_missing_data PASSED [
50%]
test_redactor.py::test_malformed_types PASSED [
75%]
test_redactor.py::test_immutability PASSED
[100%]

===== 4 passed in 0.04s
=====
```